

Microsoft Financial Performance Analysis (FY2016–FY2026)

Description

A financial analytics project analyzing Microsoft's FY2016–FY2026 financial performance using SEC EDGAR data. Python was used for data extraction, cleaning and validation; MySQL was used for structured analysis and business-question-driven queries; and Power BI was used to build an interactive management dashboard covering revenue growth, profitability, cash flow, capital investment and financial position.

Tools: Python, MySQL, Power BI

Data Source: SEC EDGAR

1. Executive Summary

This project analyzes Microsoft's financial performance over the eleven fiscal years from FY2016 to FY2026.

The objective was to assess how the company's revenue, profitability, cash generation, investment spending, and overall financial position changed over time, and to translate those findings into business-focused insights for management.

Python was used to retrieve, clean, transform, and validate the financial data.

SQL was then used to store the final analytical dataset and answer targeted business questions.

Power BI was used as the presentation layer to communicate the findings through an interactive three-page dashboard.

The analysis shows that Microsoft experienced substantial long-term revenue growth, with revenue increasing from approximately \$91.2 billion in FY2016 to \$331.8 billion in FY2026.

Profitability also improved substantially over the period. Operating margin increased from approximately 29.8% to 46.8%, while net profit margin reached 40.3% in FY2026.

However, cash flow tells a more nuanced story. Operating cash flow increased strongly, reaching approximately \$182.9 billion in FY2026, but capital expenditure also increased sharply. As a result, free cash flow declined from approximately \$74.1 billion in FY2024 to \$67.0 billion in FY2026.

The balance sheet also expanded; assets increased to approximately \$758.4 billion in FY2026, while equity increased to approximately \$442.4 billion. Equity also became a larger proportion of total assets over the period.

Overall, the analysis indicates strong long-term revenue and profitability growth, accompanied by substantially higher investment requirements. The most important management question arising from the analysis is whether the increasing capital investment is generating sufficient future economic returns.

2. Business Context

Microsoft is a large technology company with operations spanning cloud computing, productivity software, enterprise services, gaming, search, devices, and artificial intelligence.

For a company of this scale, revenue growth alone does not provide a complete picture of financial performance.

Management also needs to understand:

- whether profitability is improving alongside revenue;
- whether operating expenses are being managed efficiently;
- whether the company is converting profits into operating cash;
- how much cash is being reinvested into the business;
- whether increasing capital expenditure is affecting free cash flow;
- and how the company's financial position is changing over time.

Microsoft's FY2026 results reported revenue of \$331.839 billion, operating income of \$155.237 billion, and net income of \$133.749 billion.

This project therefore focuses on the relationship between growth, profitability, cash generation, investment, and financial position.

3. Project Objective

The primary objective was to build a financial performance analysis that answers the following business questions:

- How has Microsoft's revenue changed over time?
- Which years experienced particularly strong or weak revenue growth?
- Is profitability keeping pace with revenue growth?
- How have gross margin, operating margin, and net profit margin changed?
- Is Microsoft generating increasing amounts of operating cash?
- How much of that cash is being reinvested through capital expenditure?
- What is happening to free cash flow as investment spending increases?
- How have Microsoft's assets, liabilities, and equity changed?
- Has the composition of the company's financing changed?
- What has happened to cash and cash equivalents over time?

4. Data Extraction & Preparation

The first stage of the project was retrieving Microsoft's company facts from the SEC.

The use of SEC data was because it provided a structured source directly connected to Microsoft's regulatory financial reporting instead of a third-party financial website.

One of the most important analytical challenges was that the SEC Company Facts dataset does not simply provide one clean annual row per fiscal year.

The dataset contains:

- annual filings;
- quarterly filings;
- cumulative reporting periods;
- repeated historical values;
- different filing dates;
- fiscal-year metadata;
- and multiple observations of the same underlying financial period.

Python was used for the following;

- Retrieved the JSON
- Identified annual reporting periods
- Removed duplicate observations
- Calculated metrics
- Validated the data
- Exported the final CSV

After extracting the relevant financial concepts, the data was transformed into a single analytical table.

The final dataset contained 11 fiscal years and 22 analytical columns.

The columns were:

- fiscal_year
- revenue
- revenue_growth
- cost_of_revenue
- gross_profit
- gross_margin
- rd_expense
- sm_expense
- ga_expense
- reconstructed_opex
- operating_income
- operating_margin
- net_income
- net_profit_margin
- operating_cash_flow

- capex
- free_cash_flow
- free_cash_flow_margin
- cash_and_equivalents
- assets
- liabilities
- equity

5. Derived Financial Metrics

Several metrics were calculated to make the dataset suitable for business analysis.

- Revenue Growth = $((\text{Current Revenue} - \text{Previous Revenue}) / \text{Previous Revenue}) \times 100$

FY2016 was left as NULL because there was no previous year included in the analytical period.

- Gross Margin = $\text{Gross Profit} / \text{Revenue} \times 100$
- Operating Margin = $\text{Operating Income} / \text{Revenue} \times 100$
- Net Profit Margin = $\text{Net Income} / \text{Revenue} \times 100$
- Free Cash Flow = $\text{Operating Cash Flow} - \text{Capital Expenditure}$
- Free Cash Flow Margin = $\text{Free Cash Flow} / \text{Revenue} \times 100$

The equity ratio was calculated later in Power BI to understand how the composition of Microsoft's asset financing changed over time:

- Equity Ratio = $\text{Equity} / \text{Assets}$

6. Data Validation

Revenue calculations: Revenue growth calculations were checked against the underlying annual revenue values.

Operating expenses: Reconstructed operating expenses were compared against directly reported operating expenses.

The difference for FY2023–FY2026 was: \$0

Free cash flow: It was checked using $\text{Operating Cash Flow} - \text{CapEx}$

Balance sheet: The accounting equation was tested:

$\text{Assets} - \text{Liabilities} - \text{Equity} = 0$

7. SQL Analysis

SQL was used to query the validated dataset and answer targeted business questions, including revenue growth, profitability, cash flow, capital expenditure, and balance-sheet trends.

Revenue growth

Microsoft's revenue increased from **\$91.15B** in **FY2016** to **\$331.84B** in **FY2026**.

Highest growth: FY2022 — 17.96%

Sharp slowdown: FY2023 — 6.88%

Rebound: FY2024 — 15.67%

FY2026 — 17.79%

Finding: Revenue growth remained strong overall, despite a significant slowdown in FY2023.

Profitability

Operating margin increased from **29.83%** in **FY2016** to **46.78%** in **FY2026**.

Net profit margin also reached its highest point in **FY2026** at **40.31%**.

Gross margin increased from approximately **64.04%** in **FY2016** to a peak of **69.76%** in **FY2024** before declining to **67.94%** in **FY2026**.

FY2016: 64.04%

FY2024: 69.76% ← peak

FY2025: 68.82%

FY2026: 67.94%

Operating margin increased from **29.83%** in **FY2016** to **46.78%** in **FY2026**.

*But **FY2018** is worth flagging:*

Revenue growth: 14.28%

Operating margin: 31.77%

Net profit margin: 15.02%

Finding: Microsoft's operating efficiency improved substantially, although FY2018 had an unusual gap between operating and net profitability. This analysis does not contain enough information to establish the precise cause.

Cash Flow

Operating cash flow increased strongly over the analysis period. It reached approximately **\$182.94B** in **FY2026**.

The important finding, however, is that increasing operating cash flow did not translate into continuously increasing free cash flow.

From **FY2024** → **FY2026**:

Operating cash flow:

\$118.55B → \$182.94B

CapEx:

\$44.48B → \$115.95B

Free cash flow:

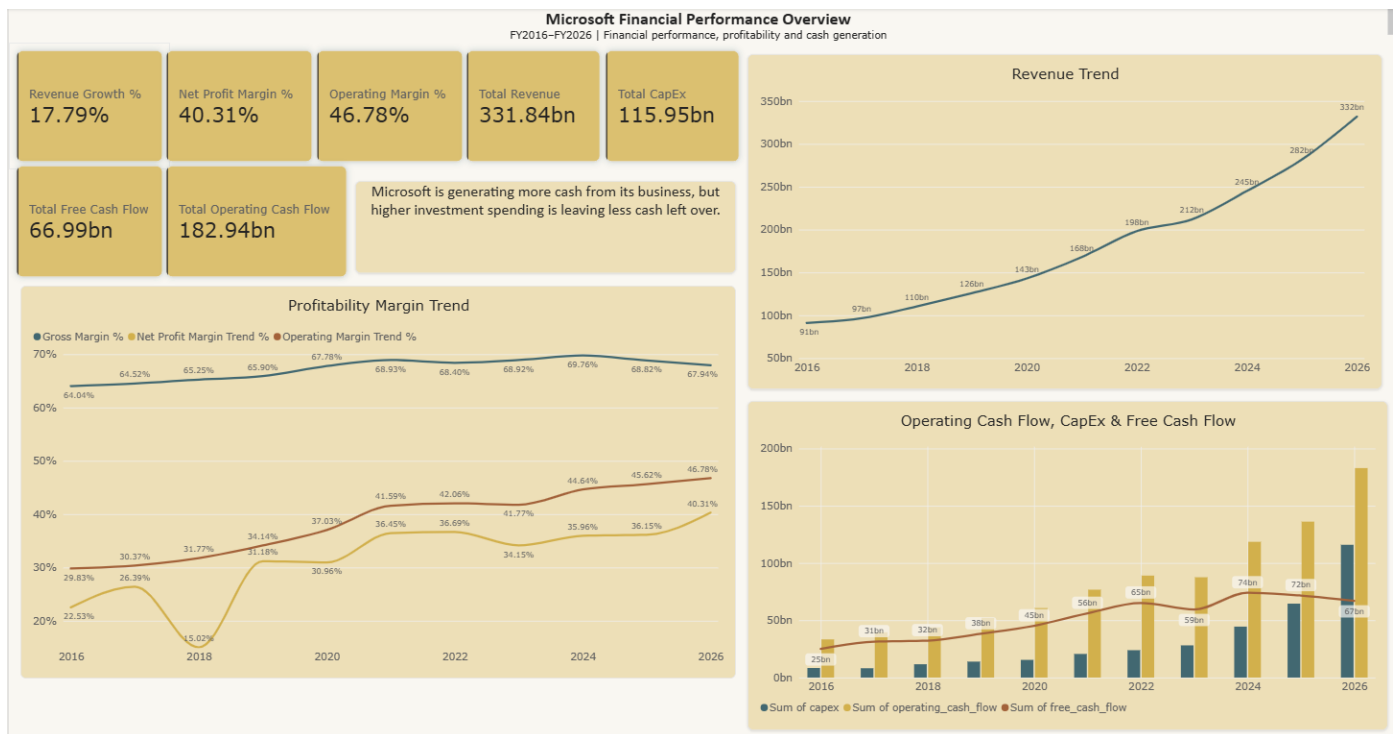
\$74.07B → \$66.99B

Finding: Microsoft generated substantially more operating cash, but rapidly increasing capital expenditure absorbed a growing share of that cash and reduced free cash flow.

8. Power BI Dashboard

The dashboard was designed on three pages.

Page 1 — Financial Performance Overview



KPIs: Revenue, Revenue Growth, Operating Margin, Net Profit Margin, Free Cash Flow, Operating Cash Flow, CapEx.

Visuals:

1. Revenue Trend

Shows the long-term increase in revenue from FY2016 to FY2026.

2. Profitability Margin Trend

Compares:

- Gross Margin
- Operating Margin
- Net Profit Margin

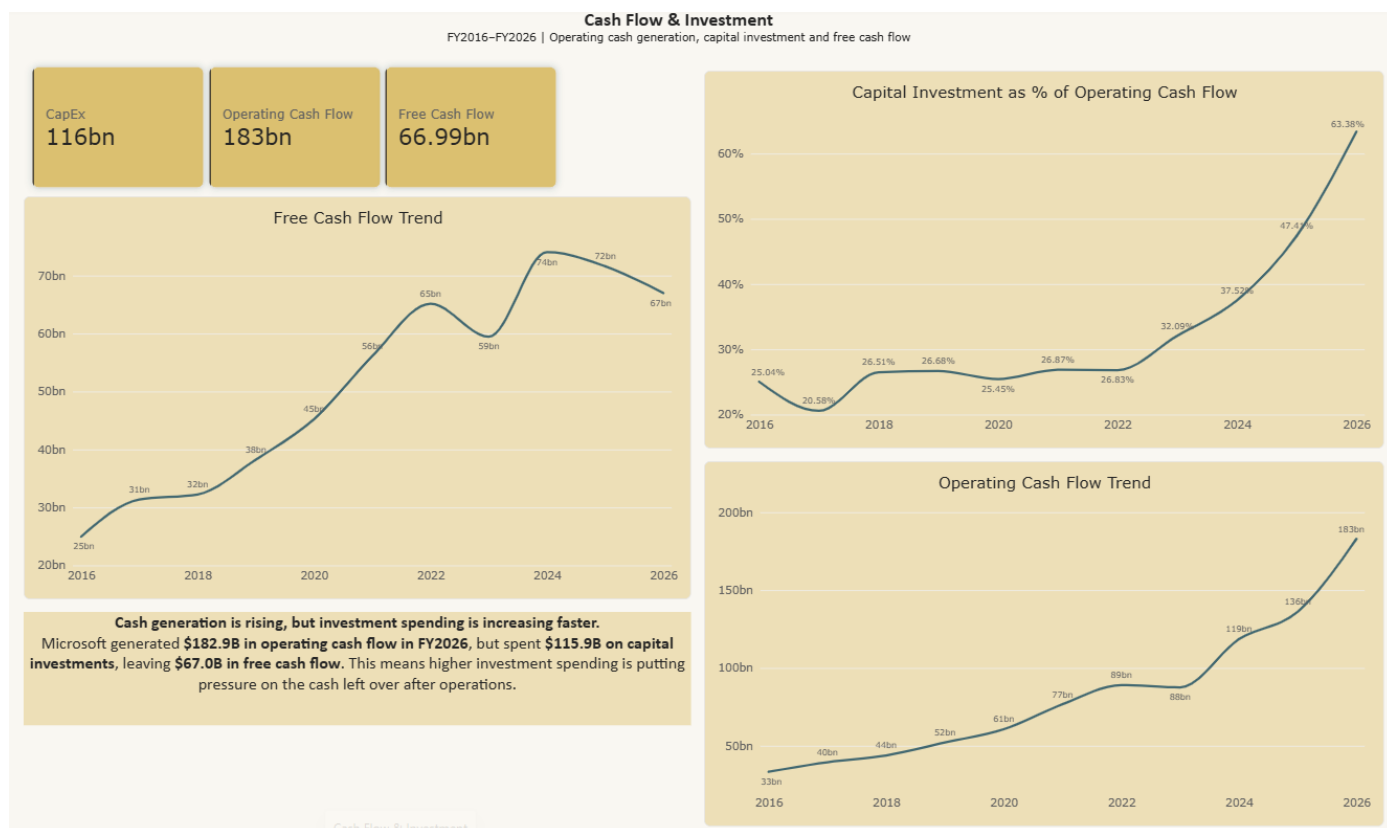
This allows management to assess whether profitability improved alongside revenue growth.

3. Operating Cash Flow vs CapEx vs Free Cash Flow

Shows the relationship between:

- cash generated from operations;
- capital investment;
- cash remaining after capital expenditure.

Insight: Microsoft is generating more cash from its business, but higher investment spending is leaving less cash available after capital expenditures.



KPIs: Operating Cash Flow, CapEx, Free Cash Flow

Visuals:

1. Capital Investment as % of Operating Cash Flow

Shows the increasing share of operating cash being consumed by capital expenditure.

2. Free Cash Flow Trend

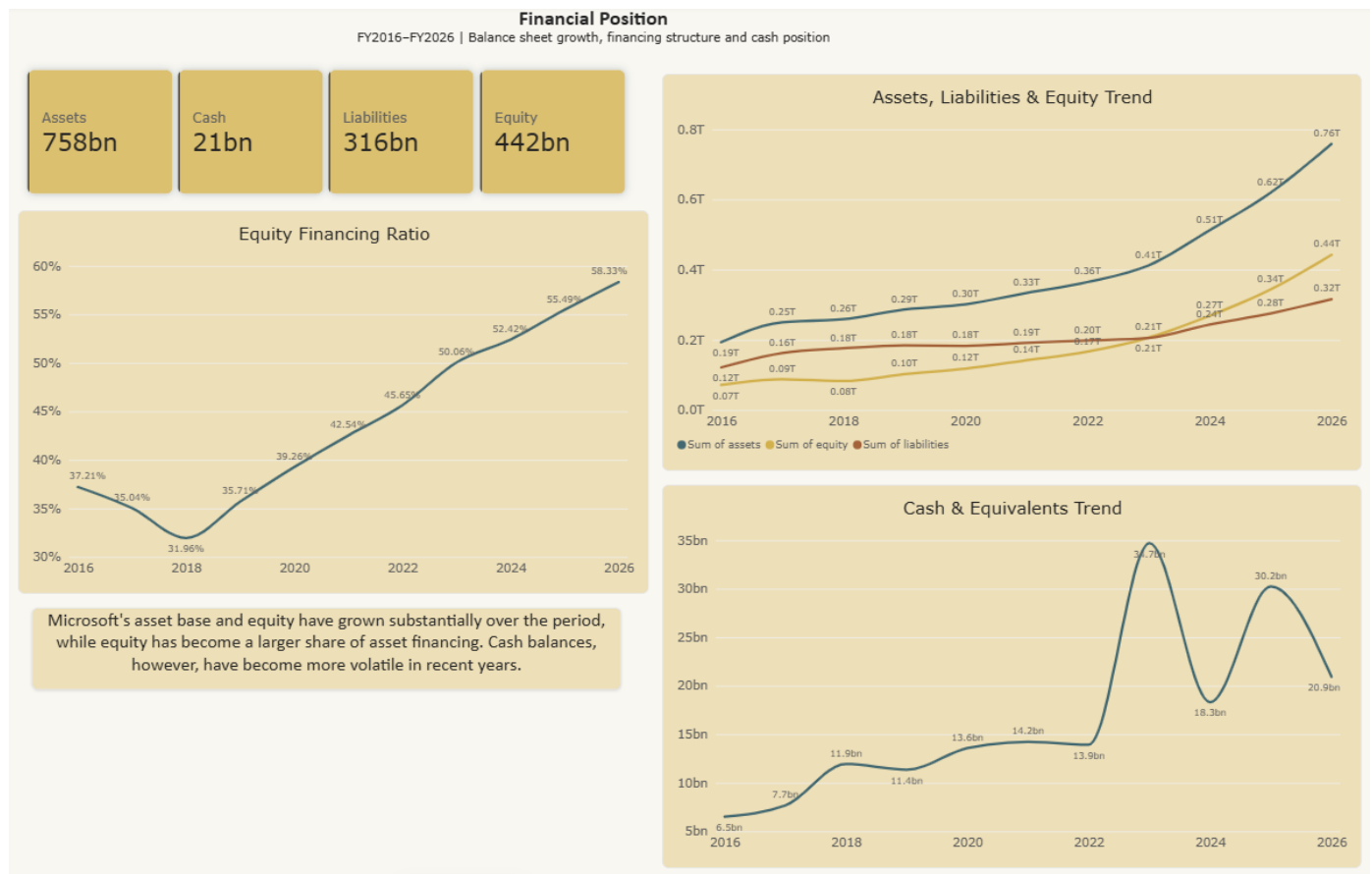
Shows the FY2024 peak followed by declines in FY2025 and FY2026.

3. Operating Cash Flow Trend

Shows that operating cash generation continues to increase despite the decline in free cash flow.

Insight: Cash generation is rising, but investment spending is increasing faster.

Page 3 — Financial Position



KPIs: Cash, Assets, Liabilities, Equity

Visuals:

1. Assets, Liabilities & Equity Trend

Shows the long-term expansion of Microsoft's balance sheet.

2. Equity Financing Ratio

Shows the increase in equity as a proportion of assets.

3. Cash & Equivalents Trend

Shows the more movement in Microsoft's cash balance, particularly from FY2023 onward.

Insight: Microsoft's asset base and equity have grown substantially over the period, while equity has become a larger share of asset financing. Cash balances, however, have become more volatile in recent years.

9. Power BI Measure Design

Several DAX measures were created specifically for the dashboard. Examples include:

- FY2026 Revenue
- FY2026 Revenue Growth
- Operating Margin
- Net Profit Margin
- Free Cash Flow
- CapEx as % of Operating Cash Flow
- Equity Ratio

The distinction between FY2026-specific KPI measures and raw historical columns was important.

FY2026-specific measures were used for KPI cards, while the raw historical fields were used for trend visuals.

10. Conclusion

Microsoft demonstrated strong financial expansion between FY2016 and FY2026.

Revenue increased substantially, while operating and net profitability also improved.

The strongest area of tension identified in the analysis is cash investment.

Operating cash generation continued to increase, but capital expenditure grew rapidly, causing free cash flow to decline from its FY2024 peak.

The company's balance sheet also expanded significantly, with equity becoming a larger proportion of total assets.

The analysis therefore presents a company with strong revenue growth and profitability, alongside a significantly higher level of investment intensity.

The key question for continued financial analysis is not simply whether Microsoft is generating more cash, but whether the substantial increase in investment, particularly capital expenditure, is translating into sufficient future growth and returns.